SOFTWARE ENGINEERING - LAB 2

Hostel Maintenance & Issue Ticketing System

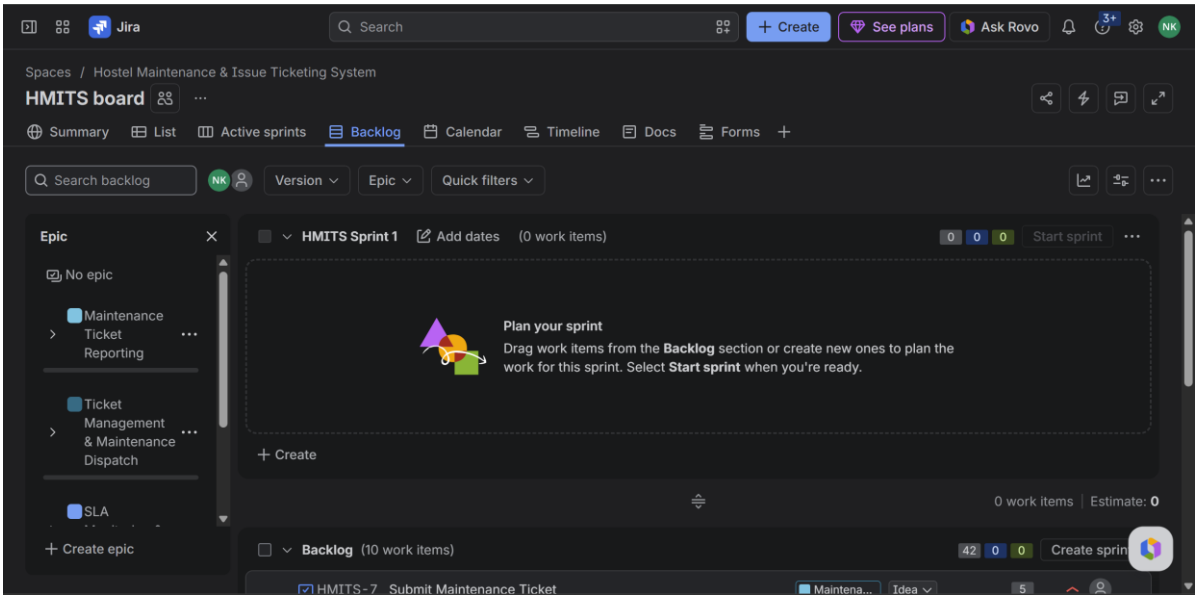
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Section: F

Problem Statement

The problem statement given here is about the Hostel Maintenance and Issue Ticketing System. The hostel management wishes to implement a ticketing system with GPS technology in order to permit the residents to raise complaints related to the plumbing, electrical and carpentry problems; whether that complaint is being sent to a maintenance team; whether the time it takes to resolve high-priority complaints is recorded in accordance with the Service Level Agreement (SLA); and whether the process of getting approval from the resident after resolution is documented. The main stakeholders of this project are the hostel residents, who create and approve tickets, and the Maintenance Warden, who deals with dispatch and escalation.

Jira Project Setup

For the Jira project, the Software Development project was initiated through the Scrum template as a Company-managed project. It was subsequently named as Hostel Maintenance & Issue Ticketing System. The board, therefore, uses HMITS as a project key, which can be seen in various identifiers that serve as issues (e.g., HMITS-3, HMITS-7).



Epic Creation

Maintenance Ticket Reporting

Covers the resident-facing activity of logging a maintenance issue, including submission, photo evidence, and issue classification.

Ticket Management & Maintenance Dispatch

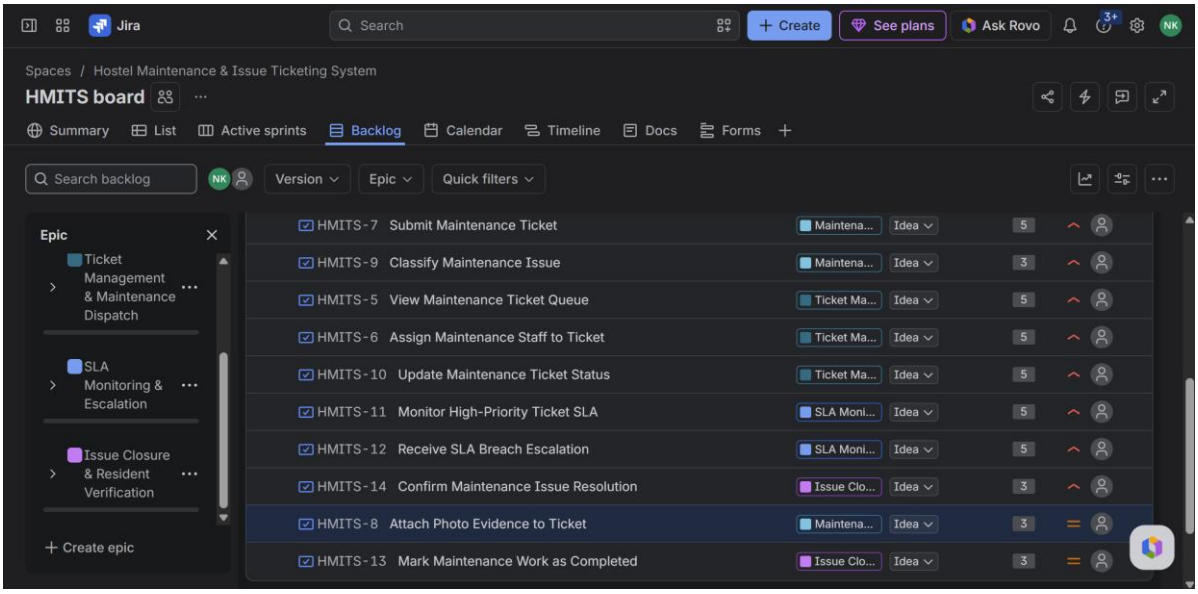
Covers warden-facing activity: viewing the ticket queue, assigning maintenance staff, and updating ticket status.

SLA Monitoring & Escalation

Covers tracking of SLA timelines on high-priority tickets and triggering escalation when an SLA is breached.

Issue Closure & Resident Verification

Covers marking maintenance work as completed and obtaining resident confirmation that the issue is resolved.



User Story Creation

For each Epic, multiple User Stories were generated in JIRA, adhering to the form of "As a [user role], I want [goal], so that [benefit]" as stated in the Lab 2 document. The detailed story description used for each story (which consists of the role/goal/benefit) is not seen in the supplied screenshots, which only show the title of the story, Epic connected, and Story Points; the table below only presents the known data obtained through JIRA.

Issue ID	User Story	Associated Epic	Story Points
HMITS-7	Submit Maintenance Ticket	Maintenance Ticket Reporting	5
HMITS-8	Attach Photo Evidence to Ticket	Maintenance Ticket Reporting	3
HMITS-9	Classify Maintenance Issue	Maintenance Ticket Reporting	3
HMITS-5	View Maintenance Ticket Queue	Ticket Management & Maintenance Dispatch	5
HMITS-6	Assign Maintenance Staff to Ticket	Ticket Management & Maintenance Dispatch	5
HMITS-10	Update Maintenance Ticket Status	Ticket Management & Maintenance Dispatch	5
HMITS-11	Monitor High-Priority Ticket SLA	SLA Monitoring & Escalation	5
HMITS-12	Receive SLA Breach Escalation	SLA Monitoring & Escalation	5
HMITS-13	Mark Maintenance Work as Completed	Issue Closure & Resident Verification	3
HMITS-14	Confirm Maintenance Issue Resolution	Issue Closure & Resident Verification	3

Ten User Stories were created in total across the four Epics, giving the workspace 14 work items overall (4 Epics + 10 User Stories), which matches the total shown in the Jira Reports summary discussed in the Results section.

Backlog Prioritization

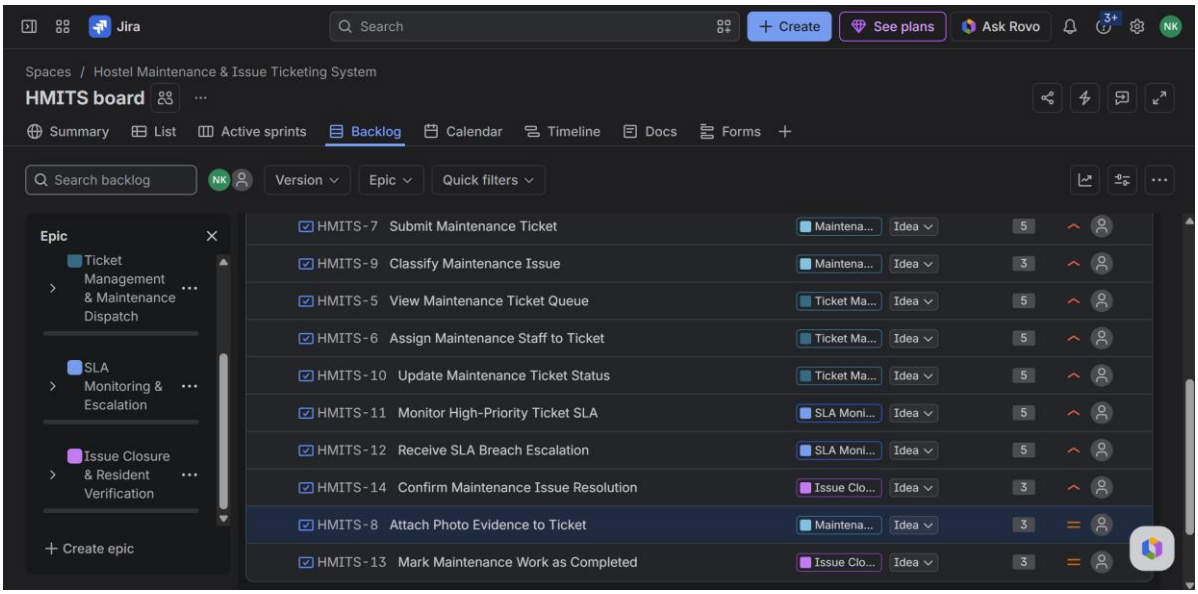
In the attached Jira screenshots, there is no visible priority field/badge, although there is an indication of the following: Epic tag, status, and Story Points for each of the items. Thus, the prioritization has to be understood either through the backlog order or through the given items selected for Sprint 1.

As mentioned in the Sprint Planning section, due to the information available in the Maintenance Ticket Reporting and Ticket Management & Maintenance Dispatch Epics, six User Stories were brought in Sprint 1 in terms of priority, while SLA Monitoring & Escalation and Issue Closure & Resident Verification stories were left for future implementation.

Story Point Estimation

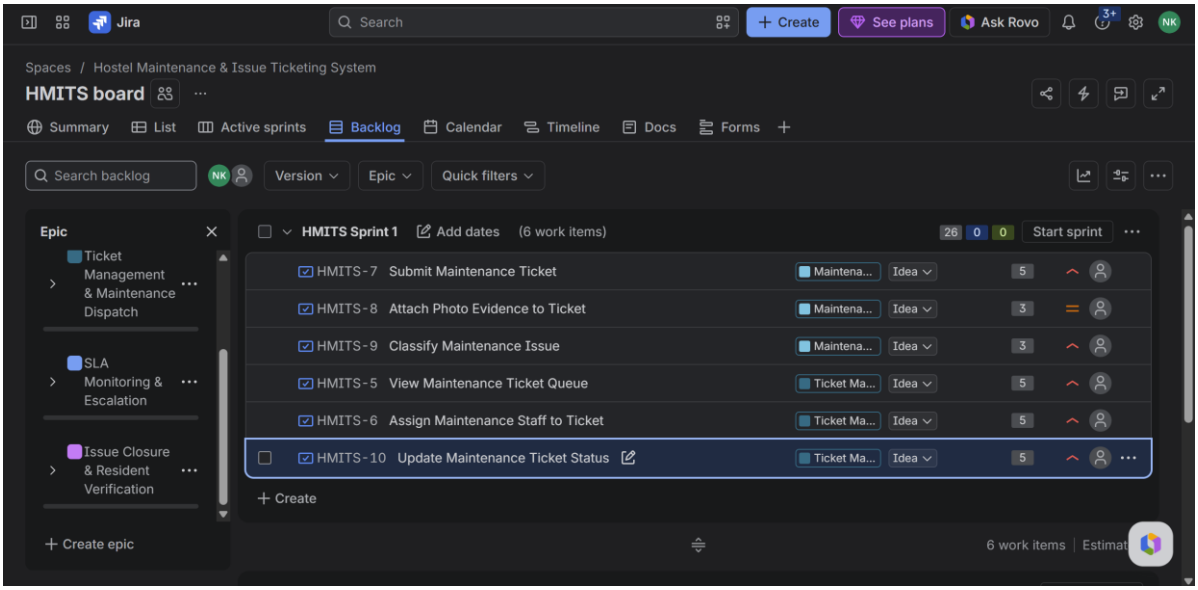
To get the value of each User Story Story Points were assigned based on the Fibonacci-based scale presented in the Lab 2 sheet (1, 2, 3, 5, 8, 13, 20, 40, 100), which indicates how much effort, complexity, and uncertainty each User Story would require. This task is an individual activity and shows planned estimates as opposed to estimates received through Planning Poker, i.e., it cannot be confirmed that any two participants took part in this Planning Poker session according to the findings in Jira.

User Story	Story Points
Submit Maintenance Ticket	5
Attach Photo Evidence to Ticket	3
Classify Maintenance Issue	3
View Maintenance Ticket Queue	5
Assign Maintenance Staff to Ticket	5
Update Maintenance Ticket Status	5
Monitor High-Priority Ticket SLA	5
Receive SLA Breach Escalation	5
Mark Maintenance Work as Completed	3
Confirm Maintenance Issue Resolution	3
Total (10 User Stories)	42

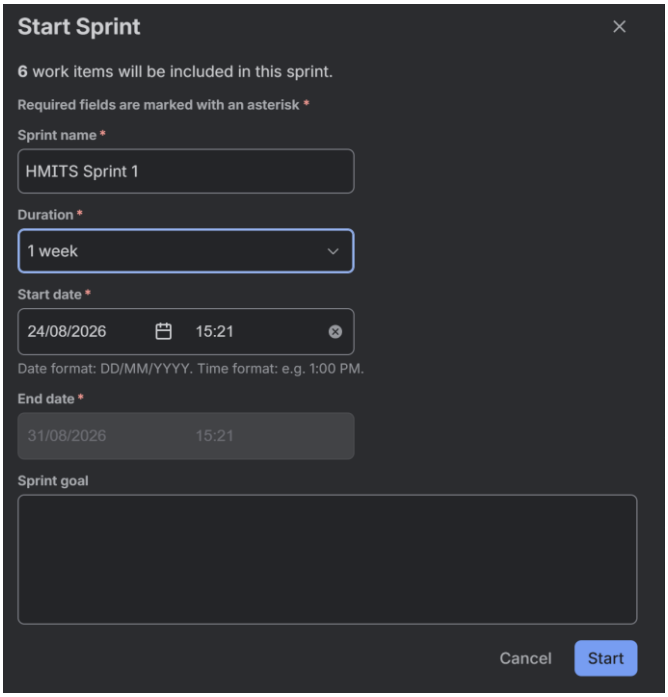


Sprint Planning and Simulation

After establishing the Epics, User Stories, priority, and Story Points, the next step was to create Sprint 1. A total of six User Stories accounting for 26 Story Points were selected from the backlog and transferred to HMITS Sprint 1. The six stories consisted of three stories under Maintenance Ticket Reporting and three stories under Ticket Management & Maintenance Dispatch. The remaining four stories, accounting for a total of 16 Story Points, which were categorized under SLA Monitoring & Escalation and Issue Closure & Resident Verification, were retained in the backlog.

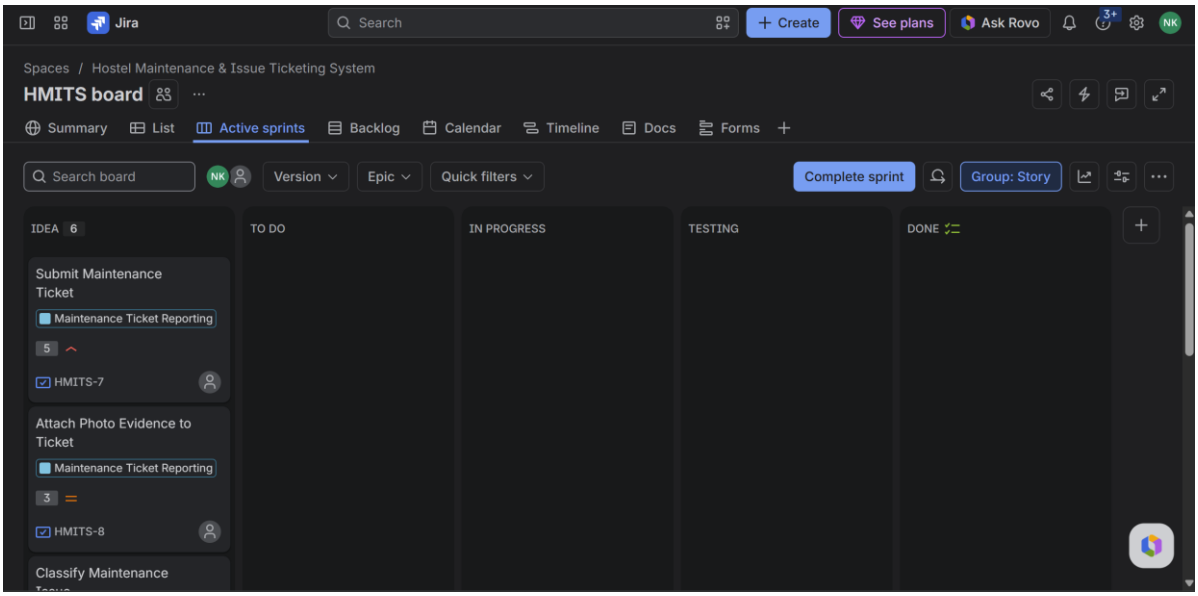


Sprint	User Stories Included	Total Story Points	Planned Duration
Sprint 1 (HMITS Sprint 1)	HMITS-7, HMITS-8, HMITS-9, HMITS-5, HMITS-6, HMITS-10	26	1 week (24/08/2026 – 31/08/2026)

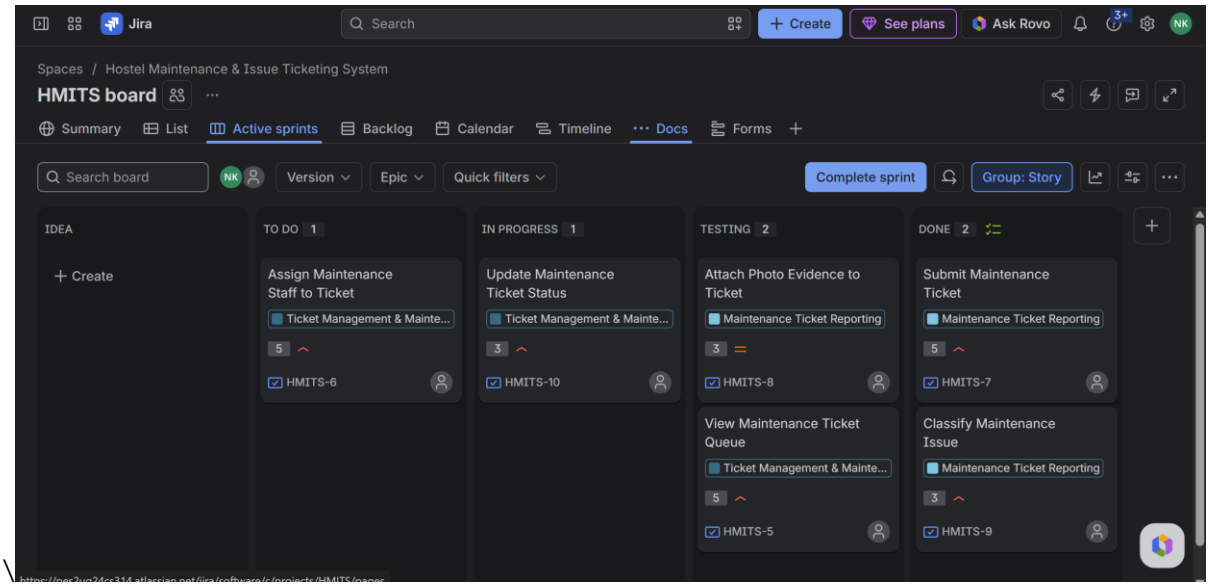


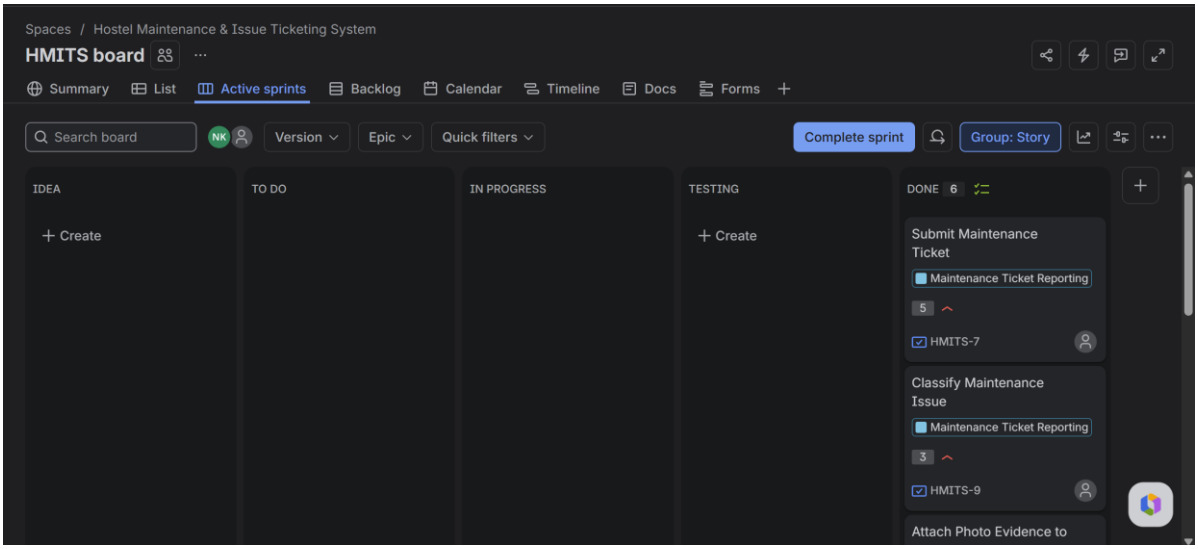
Sprint 1 Execution

The HMITS board uses a five-column workflow: Idea → To Do → In Progress → Testing → Done. Immediately after Sprint 1 was started, all six selected User Stories appeared in the Idea column.

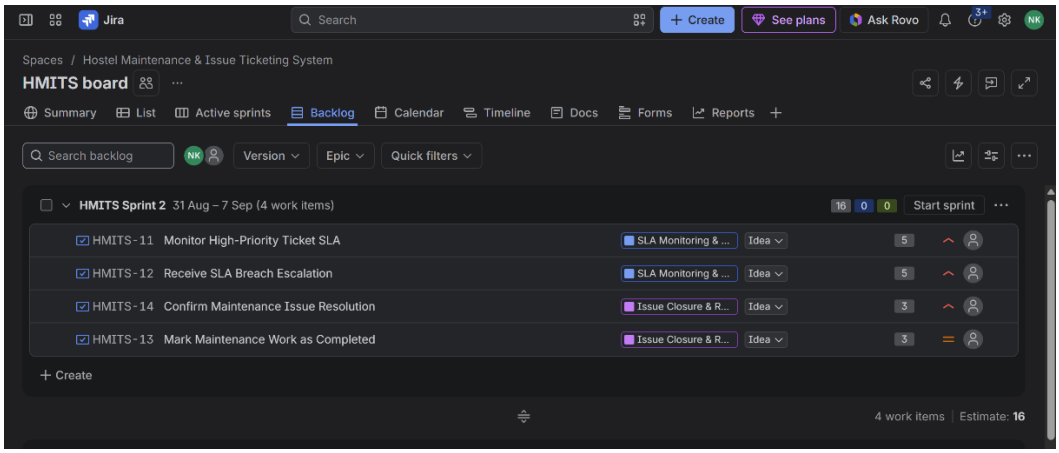


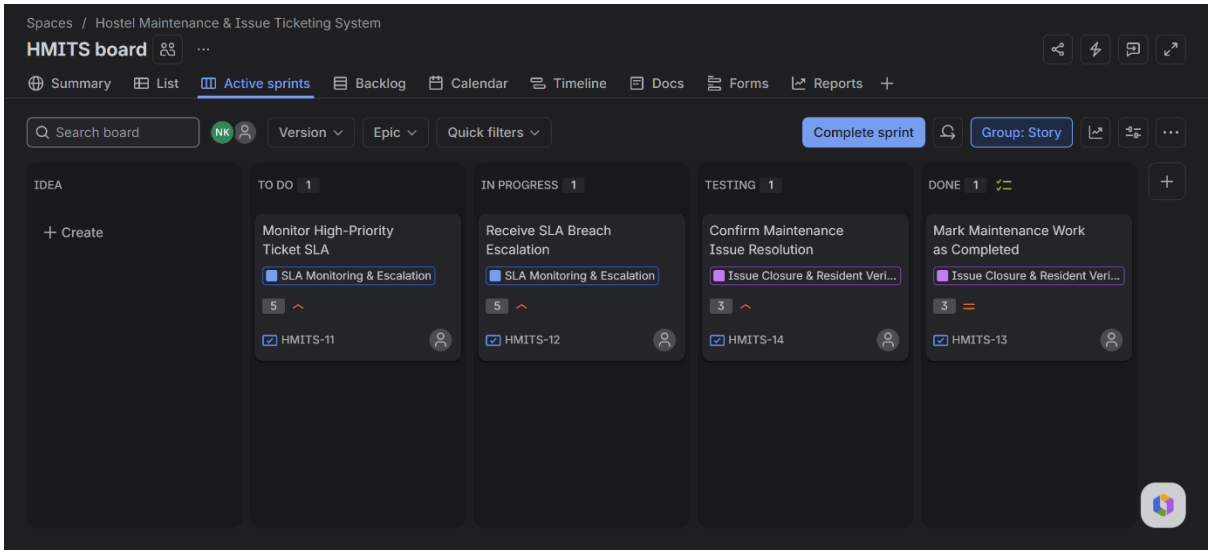
The User Stories were later gradually shifted around the board to demonstrate development progress. At one stage, HMITS-6 (Assign Maintenance Staff to Ticket) stayed in To Do, HMITS-10 (Update Maintenance Ticket Status) was in In Progress, HMITS-8 (Attach Photo Evidence to Ticket) and HMITS-5 (Get Maintenance Ticket Queue) were in Testing, while HMITS-7 (Submit Maintenance Ticket) and HMITS-9 (Classify Maintenance Issue) were moved to Done status.



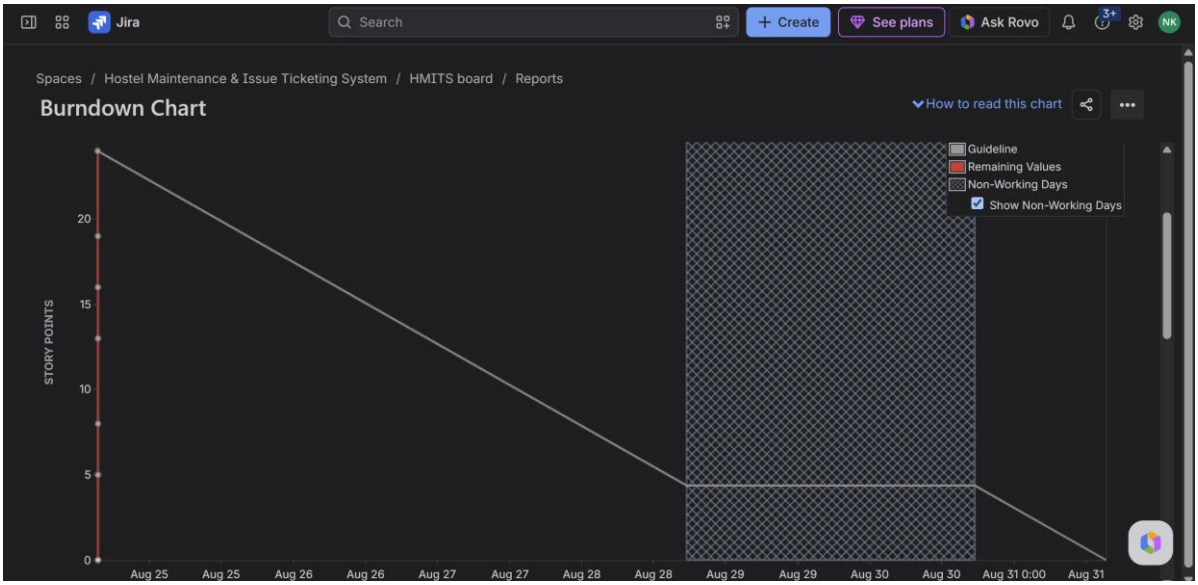


Sprint 2 Execution





Burndown Chart Analysis



Each unit of the Story Points axis on the chart increments by 5, with the Remaining Values line beginning in close proximity to the 20-count line, exactly corresponding with a total committed count of 26 Story Points for this sprint. The line starts from 24 August and slopes downwards, maintaining a flat line trend for some time around 5 units of remaining points during a period declared as non-working by Jira, only to go back down to 0 by the end date of the sprint, 31 August. This visible trend suggests that all remaining work was carried out exactly as scheduled during this sprint period, without any strange drops or spikes, during its course.

Results / Observations

The following outcomes are supported by the Jira evidence collected for this lab:

- A Company-managed Scrum project (HMITS) was created for the Hostel Maintenance & Issue Ticketing System.
- Four Epics were created, covering ticket reporting, ticket management and dispatch, SLA monitoring and escalation, and issue closure and resident verification.
- Ten User Stories were created and distributed across the four Epics, bringing the total work item count to 14, matching the Reports summary.
- The backlog was estimated using Fibonacci Story Points, for a total of 42 Story Points across all ten User Stories.
- Sprint 1 (HMITS Sprint 1) was planned with 6 User Stories totalling 26 Story Points, and run over a 1-week duration.
- Sprint 1 was simulated to completion: all 6 committed User Stories were moved through To Do, In Progress, and Testing, and reached Done.
- A Burndown Chart was generated for Sprint 1, showing remaining work reaching zero by the sprint's planned end date.

A second sprint was not evidenced in the screenshots provided; the four remaining backlog items (SLA Monitoring & Escalation and Issue Closure & Resident Verification stories) were not shown as part of an executed Sprint 2.

Reflection

Did your estimations reflect the actual effort?

Since this was a Jira sprint simulation instead of a software development sprint, no real software development hours have been compared to the Story Point estimates. Therefore, the comparison is about estimating simplicity or complexity of the work instead of comparing it to actual software development capacity. The estimated story points were proportional: simple maintenance actions such as Attach Photo Evidence to Ticket, Classify Maintenance Issue, Mark Maintenance Work Complete or

Confirm Maintenance Issue Resolution scored 3 points while more complex actions requiring interactions with other systems such as Submit Maintenance Ticket, View Maintenance Ticket Queue, Assign Maintenance Staff to Ticket, Update Maintenance Ticket Status, Monitor High-Priority Ticket SLA, and Receive SLA Breach Escalation scored 5 points. This is a fair assessment and observance of the distinction between light and heavy tasks, but it does not cover the whole Fibonacci scale range for all items in the backlog, since only two estimates with numbers 3 and 5 were used.

Was your backlog well-prioritized?

To assess if the backlog is well-prioritized, we could look at the stories chosen for Sprint 1. The six stories that have been selected are precisely those related to the core maintenance ticket process, consisting of the activity of reporting a maintenance ticket and the further processing of it up to its completion. This means that core resident-facing and dispatch functions have been prioritized over other additional functions such as monitoring SLAs and closing issues, which is a reasonable approach, as it is impossible to report a ticket or action it before the implementation of the core stories. Nevertheless, there is no priority level explicitly listed for the respective User Stories in the provided screenshots, and the only way to assess prioritization here is through the Sprint selection method.

How did your simulated sprint align with your plan?

There were 6 User Stories in Sprint 1, which was to be completed within 1 week between 24 and 31 August 2026, equalled to 26 Story Points. The analysis of the Active Sprint board confirms that the sprint was implemented in compliance with what had been planned. Indeed, all User Stories started from the Idea position to go through the columns To Do, In Progress, and Testing to reach Done. The board does not show any changes in sprint content: it is visible that no User Story had to be taken out of the scope and no new User Story was added to the sprint scope based on the illustrative screenshots.

What insights did the burndown chart give about your team's capacity?

The sprint burndown chart for Sprint 1 presents the Remaining Valuation line starting close to the total committed points and steeply going downward to zero at the end of the sprint, although remaining almost flat at the mark of five points throughout the days on which the team was not working. This pattern indicates that in this particular simulation the 26 points committed were realistic estimates of what could be achieved during the one-week sprint period, as the line reaches zero in or very close to the end of the planned time, and certainly not too early or too late. There is nothing strange in the flat part where the team was not working, since nothing could have been done on non-working days. Given that this is a one-person simulation rather than a normal sprint and a multi-person team, the burndown chart should be treated as a means of checking if the time estimation and sprint scope were realistic for the given simulation rather than assessing the velocity or capacity of a real team.